FOURIER SERIES

PERIODIC FUNCTION

A function $f(t)$ is said to be periodic if there exists a real number T such that $f(t+T) = f(t)$ for all t. Smallest such number is called period of the function.

Note: If $f(t)$ is periodic with period T, then $f(t+nT) = f(t)$ for all integers n.

If $f(x)$ is a periodic function with period $2l$ and is known in the interval $\alpha < x < \alpha + 2l$, then the Fourier series expansion of $f(x)$ in terms of an infinite sum of sines and cosines is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$$

where

$$a_0 = \frac{1}{l}\int_{\alpha}^{\alpha+2l} f(x)\,dx, \quad a_n = \frac{1}{l}\int_{\alpha}^{\alpha+2l} f(x)\cos\frac{n\pi x}{l}\,dx, \quad b_n = \frac{1}{l}\int_{\alpha}^{\alpha+2l} f(x)\sin\frac{n\pi x}{l}\,dx$$

These values of a_0, a_n, b_n are known as Euler's formulae.

Note: The graph of an even function is symmetric about $y-$axis, whereas the graph of an odd function is symmetric about the origin.

HALF RANGE SERIES

While solving various physical and engineering problems, there is a practical need for expanding functions defined over a finite range. Such an expansion is possible if functions under consideration can be extended to a periodic function which is either even or odd.

Consider a piecewise continuous function $f(x)$, defined in a finite interval $(0, l)$. Then it is possible to extend $f(x)$ to a periodic function, which is even or odd.

Consider the function $g(x)$ defined as follows:

$$g(x) = \begin{cases} f(x), & 0 < x < l \\ f(-x), & -l < x < 0 \end{cases}; \; g(x+2l) = g(x)$$

Then $g(x)$ is called an even periodic extension of $f(x)$.

The function $g(x)$ can be expanded as Fourier cosine series

$g(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi}{l}x$

where $a_0 = \frac{1}{l}\int_0^l f(x)dx$ and $a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi}{l}x\,dx$

Such an expansion of $f(x)$ is called the *half range Fourier Cosine series expansion* of $f(x)$.

Also, if $g(x) = \begin{cases} f(x), & 0 \le x \le l \\ -f(-x), & -l < x < 0 \end{cases}; \; g(x+2l) = g(x)$

Then $g(x)$ is called an odd periodic extension of $f(x)$.

The function $g(x)$ can be expanded as Fourier Sine series

$g(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi}{l}x$

where $b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi}{l}x\,dx$

Such an expansion of $f(x)$ is called the *half range Fourier Sine series expansion* of $f(x)$.

FOURIER TRANSFORMS

Consider the Fourier integral representation of the function $f(x)$ given by

$f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t)e^{is(x-t)}dt\,ds = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{isx}\left[\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)e^{-ist}dt\right]ds$

Let $F(s) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)e^{isx}dx$ ……..……. (1)

Then $f(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} F(s)e^{-isx}ds$ ……..…… (2)

The integral defined by (1) is called the Fourier transform of the function $f(x)$ and is denoted by $F\{f(x)\}$.

Given $F(s) = F\{f(x)\}$, the formula (2) defined for $f(x)$, is called the inverse Fourier transform of $F(s)$ and is denoted by $F^{-1}\{(F(s)\}$.

Note: A function $f(x)$ is said to be self-reciprocal under Fourier transforms if $F\{f(x)\} = f(s)$.

Properties of the Fourier transforms:

If $F\{f(x)\} = F(s)$, then

1. $F\{e^{iax}f(x)\} = F(s+a)$
2. $F\{f(x-a)\} = e^{isa}F(s)$
3. $F\{f(ax)\} = \frac{1}{|a|}F\left(\frac{s}{a}\right), a \neq 0$
4. $F\{\overline{f(-x)}\} = \overline{F(s)}$
5. $F\{f(-x)\} = F(-s)$
6. $F\{\overline{f(x)}\} = \overline{F(-s)}$
7. $F\{x^n f(x)\} = (-i)^n \frac{d^n}{ds^n}F(s)$
8. $F\{f^n(x)\} = (-is)^n F(s)$
9. $F\{f(x)cosax\} = \frac{1}{2}\left(F(s+a) + F(s-a)\right)$
10. $F\{f(x)sinax\} = \frac{i}{2}\left(F(s+a) - F(s-a)\right)$

PARSEVAL'S IDENTITY

If $F\{f(x)\} = F(s)$, then $\int_{-\infty}^{\infty}|f(x)|^2\,dx = \int_{-\infty}^{\infty}|F(s)|^2\,ds$

FOURIER SINE AND COSINE TRANSFORMS

Consider the Fourier cosine integral representation of a function $f(x)$

$f(x) = \frac{2}{\pi}\int_0^\infty \int_0^\infty f(t)\cos st\cos sx\ dtds, \quad x \geq 0$

$= \sqrt{\frac{2}{\pi}}\left[\sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\cos st\ dt\right]\cos sx\ ds$

Let $F_c\{f(x)\} = \sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\cos st\ dt = F_c(s)$ ……..(A)

Then $f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty F_c\{f(x)\}\cos sx\ ds$ ..……(B)

The transform $F_c\{f(x)\}$ defined by (A) is called the Fourier cosine transform of $f(x)$.The formula (B) is called the inverse Fourier cosine transform of $F_c\{f(x)\} = F_c(s)$ and is denoted by $f(x) = F_c^{-1}\{F_c(s)\}$.

Similarly, using the Fourier sine integral representation of $f(x)$ given by

$f(x) = \frac{2}{\pi}\int_0^\infty \int_0^\infty f(t)\sin st\sin sx\ dt\ ds$

we can define the Fourier sine transform of $f(x)$ denoted by $F_s\{f(x)\}$ as

$F_s\{f(x)\} = \sqrt{\frac{2}{\pi}}\int_0^\infty f(t)\sin st\ dt = F_s(s)$

Then the inverse Fourier sine transform of $F_s(s)$ is defined as

$f(x) = F_s^{-1}\{F_s(s)\} = \sqrt{\frac{2}{\pi}}\int_0^\infty F_s(s)\sin sx\ ds$

Note: A function $f(x)$ is said to be self-reciprocal under Fourier cosine (sine) transform if $F_c\{f(x)\} = f(s)$ $\left(F_s\{f(x)\} = f(s)\right)$.

Properties of Fourier cosine/sine transforms:

1. Both Fourier cosine and sine transforms are linear
   $F_c\{c_1f(x) + c_2g(x)\} = c_1F_c\{f(x)\} + c_2F_c\{g(x)\}$
   $F_s\{c_1f(x) + c_2g(x)\} = c_1F_s\{f(x)\} + c_2F_s\{g(x)\}$ where $c_1$and c_2 are constants.
2. $F_c\{f(x)cosax\} = \frac{1}{2}\left(F_c(s+a) + F_c(s-a)\right)$
   $F_c\{f(x)sinax\} = \frac{1}{2}\left(F_s(s+a) - F_s(s-a)\right)$
3. $F_s\{f(x)cosax\} = \frac{1}{2}\left(F_s(s+a) + F_s(s-a)\right)$

$F_s\{f(x)sinax\} = \frac{1}{2}\big(F_c(s-a) - F_c(s+a)\big)$

4. $F_c\{f(ax)\} = \frac{1}{a}F_c\left(\frac{s}{a}\right)$

   $F_s\{f(ax)\} = \frac{1}{a}F_s\left(\frac{s}{a}\right)$

5. If $f(x) \to 0$ as $x \to \infty$, then

   $F_c\{f'(x)\} = -\sqrt{\frac{2}{\pi}}f(0) + sF_s(s)$ and $F_s\{f'(x)\} = -sF_c(s)$

6. $F_c\{f''(x)\} = -\sqrt{\frac{2}{\pi}}f'(0) - s^2F_c(s)$ and $F_s\{f''(x)\} = \sqrt{\frac{2}{\pi}}sf\ (0) - s^2F_s(s)$ provided $f(x)$ and $f'(x) \to 0$ as $x \to \infty$

7. $F_c\{xf(x)\} = \frac{dF_s}{ds}$ and $F_s\{xf(x)\} = -\frac{dF_c}{ds}$

8. If $F_c\{f(x)\} = F_c(s), F_c\{g(x)\} = G_c(s), F_s\{f(x) = F_s(s)$ and $F_s\{g(x)\} = G_s(s)$ exist, then $\int_0^\infty F_c(s)G_c(s)ds = \int_0^\infty F_s(s)G_s(s)ds = \int_0^\infty f(x)g(x)dx$ and $\int_0^\infty |F_c(s)|^2\,ds = \int_0^\infty |F_s(s)|^2\,ds = \int_0^\infty |f(x)|^2\,dx$ which is called Parseval's Identity.

HARMONIC ANALYSIS

Let f be an unknown periodic function given in the interval $[a, b]$ with period $2l$ such that

x	$a = x_0$	x_1	x_2	...	$b = x_N$
$f(x)$	y_0	y_1	y_2	...	y_N

where x_i are equally spaced by h $\left(\frac{b-a}{N} = \frac{2l}{N}\right)$ units in $[a, b]$. Then

$$a_0 = \frac{1}{l}\int_a^b f(x)dx = \frac{2}{N}\sum_{i=1}^{N} y_i$$

$$a_n = \frac{1}{l}\int_a^b f(x)\cos\left(\frac{n\pi x}{l}\right)dx = \frac{2}{N}\sum_{i=1}^{N} y_i\cos\left(\frac{n\pi x_i}{l}\right)$$

$$b_n = \frac{1}{l}\int_a^b f(x)\sin\left(\frac{n\pi x}{l}\right)dx = \frac{2}{N}\sum_{i=1}^{N} y_i\sin\left(\frac{n\pi x_i}{l}\right)$$

Note:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

i.e.,

$$f(x) = \frac{a_0}{2} + a_1 \cos\left(\frac{\pi x}{l}\right) + b_1 \sin\left(\frac{\pi x}{l}\right) + a_2 \cos\left(\frac{2\pi x}{l}\right) + b_2 \sin\left(\frac{2\pi x}{l}\right) + \cdots$$

$a_1 \cos\left(\frac{\pi x}{l}\right) + b_1 \sin\left(\frac{\pi x}{l}\right)$ is the first harmonic and $\sqrt{a_1^2 + b_1^2}$ is the amplitude of the first harmonic

$a_2 \cos\left(\frac{2\pi x}{l}\right) + b_2 \sin\left(\frac{2\pi x}{l}\right)$ is the second harmonic and $\sqrt{a_2^2 + b_2^2}$ is the amplitude of the second harmonic

MATRIX DECOMPOSITION

EIGENVALUES AND EIGENVECTORS

Let A be a square matrix of order n. A scalar λ is an *eigenvalue* of A if there is a non-zero column vector v of length n such that $Av = \lambda v$. The vector v is called an *eigenvector* of A corresponding to the eigenvalue λ.

Equivalently, a scalar λ is an eigenvalue of A if $\det(A - \lambda I_n) = 0$, and the eigenvectors corresponding to λ are the solutions of the homogeneous system $(A - \lambda I_n)X = 0$.

The characteristic polynomial of A is denoted by $\Delta(A)$, and is defined as $\Delta(A) = \det(A - \lambda I_n)$. Henceforth, the eigenvalues of A are the roots of the characteristic polynomial.

If λ is an eigenvalue of A, then the solution space of the system $(A - \lambda I_n)X = 0$ is called the *eigenspace* of A corresponding to the eigenvalue λ and is denoted by $E_\lambda(A)$.

Note: Let A be a square matrix of order n. Let S_k denote the sum of all principal minors of A of order k. Then $\Delta(A) = \lambda^n - S_1\lambda^{n-1} + S_2\lambda^{n-2} - \cdots + (-1)^n S_n$ where S_k is the sum of all principal minors of A of order k.

DIAGONALIZATION OF A MATRIX

Square matrices A and B are *similar* if there exists an invertible matrix P such that $A = PBP^{-1}$

A square matrix A is *diagonalizable* if it is similar to a diagonal matrix

Note: A square matrix A of order n is diagonalizable if and only if it has n linearly independent eigenvectors. If a square matrix A of order n has n distinct eigenvalues, then it is diagonalizable.

SPECTRAL THEOREM FOR SYMMETRIC MATRICES

A square matrix A is *orthogonally diagonalizable* if there exists an orthogonal matrix P such that $A = PDP^T$, where D is a diagonal matrix. The following is the spectral theorem for symmetric matrices. Let A be a symmetric matrix of order n. Then

i. A has n real eigenvalues, counting multiplicities

ii. If λ is an eigenvalue of A with multiplicity k, then the dimension of eigenspace $E_\lambda(A)$ is k

iii. The eigenvectors corresponding to different eigenvalues are orthogonal

iv. A is orthogonally diagonalizable

CHOLESKY DECOMPOSITION

Let $A = [a_{ij}]_{n\times n}$ be a positive definite matrix. Then A can be written as $A = LL^T$ where $[l_{ij}]_{n\times n}$ is a lower triangular matrix and it is called as Cholesky decomposition of A. The entries of the matrix L can be computed as follows:

$$l_{11} = \sqrt{a_{11}}$$

$$l_{j1} = \frac{a_{j1}}{l_{11}},\ j = 2, \dots, n$$

$$l_{jj} = \sqrt{a_{jj} - \sum_{S=1}^{j-1} l_{js}^2},\ j = 2,3, \dots, n$$

$$l_{pj} = \frac{1}{l_{jj}}\left[a_{pj} - \sum_{S=1}^{j-1} l_{js}\, l_{ps}\right],\ p = j+1,\ j+2, \dots,\ n \text{ and } j \geq 2$$

LU DECOMPOSITION

Representation of a square matrix A as a product of lower triangular matrix L and upper triangular matrix U, that is, $A = LU$ is called a LU decomposition of matrix A. Note that an invertible matrix A has LU decomposition if and only if all its leading principle minors are non-zero.

QR DECOMPOSITION

The QR decomposition of a real square matrix is $A = QR$ where Q is an orthogonal matrix and R is an upper triangular matrix.